A codebase goes through different stages in its journey from design to deployment in production:

- *Design stage*—Technologies, dependencies, and tools needed by a specific application feature are decided.
- *Build stage*—The codebase is compiled and packaged together with its dependencies as an immutable artifact called a *build*. The build artifact must be uniquely identified.
- *Release stage*—The build is combined with a specific configuration for the deployment. Each release is immutable and should be uniquely identifiable, such as by using semantic versioning (for example, `3.9.4`) or a timestamp (for example, `2022-07-07_17:21`). Releases should be stored in a central repository for easy access, like when a rollback to a previous version is required.
- *Run stage*—The application runs in the execution environment from a specific release.

The 15-Factor methodology requires a strict separation of these stages and doesn't allow changes to the code at runtime, since that would result in a mismatch with the build stage. The build and the release artifacts should be immutable and labeled with a unique identifier to guarantee reproducibility.

2.1.5 Configuration, credentials, and code

The 15-Factor methodology defines configuration as everything likely to change between deployments. Whenever you need to change the configuration for an application, you should be able to do so without any changes in the code, and without building the application again.